Examiner reports

Q2.

The use of the trapezium rule was well understood with a majority of students presenting a fully correct solution to part (a) with the final answer given to the degree of accuracy requested. Students need to take extra care when transferring correct decimal values from a table into the formula; this was particularly relevant in this series to those values which had leading zeros, namely for f(4) and f(5). The most common method error was to use end values 0 and 4 for x instead of the correct end values 1 and 5. A small minority of students left their final answer in the form $\frac{694}{1105}$ which resulted in the loss of the final accuracy mark. In part (b)(i), most students were able to integrate one of the terms correctly but it was not uncommon to see the integral of $x^{-\frac{3}{2}}$ resulting in a positive coefficient or an incorrect numerical simplification of 1/–0.5 . Very few students failed to score the method mark in part (b)(ii) for being able to deal with limits in a correct manner.

Q3.

The trapezium rule question in part (a) was generally well answered with the majority of students taking care to include sufficient brackets and rounding their final answers to the required degree of accuracy. In part (b), many students correctly recognised the transformation as a stretch and a majority of these stated its correct direction. However, only the most able students stated the correct scale factor as 2 with most other students incorrectly stating it as either 8 or $\frac{1}{8}$. Although fully correct solutions to part (c) were rare, a significant number of students did gain some credit. The most common approach was to first find an expression for g(x) before substituting the value 4 for x. Other than sign errors, the most common error was to write g(x) = $\sqrt{(x-2)^3+1-0.7}$ or its equivalent. Those who gave the correct expression, $\sqrt{(x-2)^3+1}$ – 0.7, usually went on to score all 3 marks. The other approach, attempting to find and use the original point to transform, was rarely seen.

Q4.

This question, which tested the use of the trapezium rule, was again a good source of marks for most candidates. In part (a), the two most common mistakes were treating $\frac{2^x}{x+1}$ as $\frac{2x}{x+1}$ or failing to write the final answer to three significant figures, as illustrated by $\frac{193}{30}$. It was rare to find a candidate who could not recall how a better approximation to the value of the integral could be obtained using the trapezium rule.

Q5.

The application of the trapezium rule in part (a), followed by the request for the vector of a translation in part (b), provided a good opportunity for most students to make a good start to this final question. Using log laws in the correct order on log($10x^2$) defeated the average and below average students, even with the answer given. It was quite common to see

$\log(10x^2)$ initially written as $2\log(10x)$ and then, with less than convincing logic, the incorrect expression became the right-hand side of the printed result. Identifying and describing the stretch was, as expected, discriminating. Only a small minority identified $f(ax)$ as $f(\sqrt{10}\,x)$ here, although many more gained some partial credit. Having this part (c)(ii), following the given result in part (c)(i), was a deliberate connection often ignored by students. Again, in answering part (c)(iii) students could make good use of the printed result, and this was frequently taken up by better students who generally made good progress for the first two or three marks. However, in general, only the most able students could give the correct gradient of OP in the form requested.

Q6.

The majority of candidates presented a fully correct trapezium rule, working to the required degree of accuracy and so scored full marks. Common errors were normally arithmetical with relatively few candidates failing to score the method mark. In part (b), most candidates indicated that they knew that a multiple of 3 was needed somewhere, although fully correct solutions were

rare. The most common error was to replace x^3 by $\frac{1}{3}x^3$ rather than by $\left(\frac{1}{3}x^3\right)^3$. Another, more serious, error was to multiply the given expression by 3 and write $g(x) = 3\sqrt{27x^3+4}$ or some other variation of this error, for example, $g(x) = \sqrt{81x^3+12}$.

Q7.

Part (a) was answered well with most candidates using the trapezium rule with the correct number of strips and values substituted correctly. The most common errors were using degrees instead of radians and incorrect use of brackets.

Most candidates recognised that the required transformation for part (b) was a stretch, but were less sure of the direction and/or scale factor.

In part (c) the most common approach was to attempt to write the given equation in terms of
tan x with varying degrees of success. Although some very good solutions were seen, a common wrong method is illustrated by '2tan x = 1, $2x$ = 0.785, 3.93,….. x = 0.393, 1.97, …..'

Those who correctly reached either $\sin^2 x = \frac{1}{5}$ or $\cos^2 x = \frac{4}{5}$ rarely scored more than one of the
four marks because they forgot to consider both the positive and negative roots of these equations.

Q8.

Candidates' sketches generally displayed a thorough understanding of the graph of $y = 2^x$. In part (b)(i) the trapezium rule was frequently applied correctly and a smaller proportion of candidates than normal failed to give their final answer to the stated degree of accuracy. Part (b)(ii) was almost always answered correctly. In part (c), the question asked for a description of a geometrical transformation. Although answers which gave two separate translations were condoned, the penalty for descriptions which involved two different transformations, for example stretch and translation, was severe. However many correct descriptions were presented, which indicates the continued improvement in candidates' performance on this topic. Part (d), as expected, proved to be demanding. The common wrong approach is illustrated by 'log y = log 2^{x+k} + log 3'. Better candidates applied the correct substitution and manipulation to reach $2^k = 5$, but a significant minority of these candidates, after taking logs and using the log law, left their answer in the form $k = \frac{\log 5}{\log 2}$.

Q9.

Parts (a) and (b) gave the opportunity for all to pick up marks, with the majority giving the two decimal place answer required in part (b). Many candidates scored just one of the two marks in part (c). The translation approach was the more popular, but the fact that the vector needed to be $\begin{bmatrix} \frac{3}{4} \\ 0 \end{bmatrix}$ was too subtle for most, as was the scale factor of $\frac{1}{8}$ for those using a stretch approach. Similarly in part (d), needing $4(x - 1)$ rather than $4x - 1$ defeated the majority. However, this caused no undermining of confidence as many answered part e(i) well, with sufficient justification to support the given answer.

It should be noted, however, that answers which simply started with $k = \frac{8 \times 5}{4}$ with no evidence of using correct log laws could not earn credit. In part (e)(ii) most candidates knew they had to solve '$\frac{5}{4} = 2^{4x-3}$' and started by taking logs, but then resorted to calculators rather than establish an exact result (with numbers hinted at in the previous part). The best solution, given by some, was to multiply both sides by '2^3' and then take logs, to the base 10, of '$10 = 2^{4x}$', which led directly to the result.

Q10.

The trapezium rule was generally well understood with very little evidence of candidates mixing up 'ordinates' and 'strips'. However some candidates failed to gain the final mark in part (a) because they did not give their final answer to the three significant figures asked for. Candidates were required to show some evidence of correct evaluations of their surds before the possible award of the two accuracy marks. Although in part (b) the vast majority of candidates correctly stated how they could obtain a better approximation to the value of the integral using the trapezium rule, a common wrong answer was 'use more than three significant figures'.

Q11.

The trapezium rule was again generally well understood with very little evidence of candidates mixing up 'ordinates' and 'strips'. Fewer candidates than normal failed to give their final answer to the required number of significant figures although there was some evidence of premature approximation/wrong rounding in a minority of solutions.

Part (b), as expected, caused more problems. Generally weaker and average candidates did not understand what was required and many better candidates failed to use brackets sensibly and gave the expression for f(x) as ‘ $\sqrt{2x^3} + 1$ ’ for which they were awarded 1 mark.

Q12.

The trapezium rule was usually well understood, with far fewer candidates than in previous years mixing up “ordinates” and “strips”. In general, candidates’ answers on this topic were better than in previous sessions.

Q13.

Candidates generally applied the trapezium rule correctly and gave the final answer to the given degree of accuracy. Many candidates failed to gain full marks in part (b)(ii) as their diagrams did not show four trapeziums. There were also a significant number of candidates who either failed to draw any diagram or who drew the sloping sides of their trapeziums under the curve. Most candidates realised that the geometrical transformation was a stretch, but the wrong answer ‘stretch parallel to the y-axis, scale factor 3’ appeared almost as frequently as the correct answer. A high proportion of candidates showed that they were able to use logarithms to solve the equation $6^{3x} = 84$ but some only gave their answer to two decimal places. The final part of the question was poorly answered, although a minority of candidates did score some credit for partially correct answers, for example, f(x) = $6^{x+1} - 2$.

Q14.

The trapezium rule was usually well understood, with fewer candidates mixing up ‘ordinates’ and ‘strips’; although failing to use sufficient brackets correctly is still a problem. Those who used $2x$ instead of 2^x could score a maximum of two marks for the question.

Q15.

This question also proved to be a good source of marks for a majority of candidates. Normally part (a) was answered correctly but there were also signs of poor examination technique. Candidates who just wrote down the y-coordinate of P as ‘3’ scored no marks but those who wrote ‘$3(2^0 + 1) = 3$’ scored 1 of the 2 marks available. Responses to the trapezium rule question were mixed with many candidates scoring full marks but a greater number of candidates failing to pick up even the method mark. The two main method errors involved either the use of seven ordinates rather than the required four or the use of $x = 0, 1, 2, 3$ only.

In part (c)(i) a significant number of candidates were unable to gain the mark because they wrote 3×2^x as 6^x. Many candidates gained the 2 method marks for using logarithmic rules but a significant minority failed to round 2.5849 correctly to three significant figures. The wrong answer 2.59 was seen more than expected.

Q16.

Many candidates had a clear understanding of how to apply the trapezium rule but careless
non-use of sufficient brackets resulted in a significant number of incorrect solutions. Other

common errors were (i) to use $h = \frac{b-a}{n} = \frac{4-0}{5}$ = 0.8 which suggests that the meaning of n in the relevant section of the formulae booklet was not fully understood (ii) to assume y_0 equals 0.

A number of candidates lost the final accuracy mark because they gave their final answer as 1.3294 or 1.33 Part (b) was generally answered well although some failed to read the question carefully enough and gave the incorrect answer 'Use integration'.

Q17.

This question proved to be a good source of marks for many candidates. Most candidates answered both parts of (a) correctly although the usual errors, $3^\circ = 0$ or $3^\circ = 3$ and $3^X = 27$ $\Rightarrow x = \sqrt[3]{27}$ were seen. The trapezium rule, required in part (b), was generally well understood although some candidates are still mixing up 'ordinates' and 'strips' or failing to use sufficient brackets correctly. Although many candidates scored well in parts (c) there were others in (i) who used the relevant law of logarithms incorrectly or others who failed to indicate that logarithms had been used at all. In (ii) not all candidates realised that a simple substitution of 13 for 3^X in $k = 27 - 3^X$ with evaluation was all that was required. Descriptions of transformations continue to pose a problem for candidates. Many candidates gained partial credit for their sketches but both marks were rarely awarded.

Q18.

Many candidates incorrectly evaluated $(3^0 + 1)$ as 1 in part (a) although surprisingly they frequently recovered and used the correct answer, 2, in part (b). In general, the trapezium rule was well understood, although it was relatively common to find an arithmetical slip in evaluating $(3^x + 1)$; the '1' was not added in one of the terms. Candidates should have given the final answer to the degree of accuracy specified in this numerical integration question. Some of the weaker candidates just integrated $(3^x + 1)$ directly as $(3^{x+1} + x)$ The use of a diagram to show that the approximation is an overestimate was well understood, although some used rectangles rather than the required trapeziums and others had the sloping sides of their trapeziums in the wrong positions, that is, below the curve. Part (c) was generally well answered although some weaker candidates tried to take logarithms of each term rather than to rearrange the equation into the form '$3^x = 4$' as a first step. Those candidates who used trial and improvement usually failed to produce an acceptable solution. In part (d), a common wrong expression for f(x) was "$-3^x + 1$", although a significant minority did obtain the correct answer " $3^{-x} + 1$".

Q19.

This question proved to be a good source of marks for many candidates. In part (a)(i) most candidates quoted the correct value for the power p. Although many candidates produced correct answers for the integration, all the usual errors, as illustrated by $\frac{3}{2}x^{1.5}$, $x^{1.5}$, $\frac{x^{-0.5}}{-0.5}$, and $\frac{\sqrt{x}^2}{2}$, were seen. Most candidates stated and used the correct limits in part (a)(iii) but evaluation was sometimes poor. Many of the candidates found a correct equation for the tangent but it was surprising to find a significant minority of these candidates either making no attempt to find the area of triangle AOB or giving the area as a negative value. For full marks in part (c) the examiners were expecting to see the word ‘translation’. Some candidates wrote ‘tr’ but this was not considered to be clear enough to distinguish between ‘transformation’ and ‘translation’. Common wrong answers involved ‘stretch’ and vectors in the wrong direction. In part (d) there was a significant minority of candidates starting their solution with the incorrect statement $\sqrt{x-1} = \sqrt{x} - \sqrt{1} = \sqrt{x} - 1$ but in general the trapezium rule was well understood and many candidates were able to give the final answer to the correct required degree of accuracy in this numerical integration question.